Supported kernels and finite two-group quotients

Human Compression Project

Abstract

We study marked groups through the intersections of subgroups generated by subsets of their markers. A construction over a group with deletion maps attaches local extensions whose kernels are invisible on proper local subgroups. We prove generation of the full compatible product, compute every parabolic kernel and obtain the intersection property. The same coordinates give local recognition, deletion maps, lower-central identities, separator factorizations and transfer of the flag property for coset geometries. For finite two-groups whose noncommuting diagram is a forest, we identify a canonical class-two quotient and a finite, exhaustive parameter space of marked quotients of any prescribed excess order. Explicit affine square groups supply local models of every required two-power order. The paper includes the supporting quotient, Frattini and amalgam arguments so that subsequent applications can cite complete statements with specified hypotheses.

1 Introduction

A presentation by involutions does not by itself determine a regular polytope or hypertope. The distinguished generators must have the prescribed pair orders, their generated subgroups must intersect correctly, and a geometric conclusion may require a further common-representative condition on cosets. Establishing these properties separately for every presentation can obscure the part of the proof that is independent of its numerical parameters.

The main construction here retains an already understood marked quotient and records the extra relations in separate kernel coordinates. Each attached group is required to agree with the quotient on every proper part of its support. This hypothesis lets us isolate its whole kernel, even when one support contains another. Once the kernels have been isolated, all marked subgroup intersections and orders follow from a single coordinate calculation. The geometric assertion uses an additional proved transfer of typed coset Helly; it is not inferred from the intersection property alone.

A second branch identifies the quotient which every finite two-group marking on a forest must have. The proof uses the intersection property in the original group, a descending lower-central argument and the separation of a forest at an edge. It then reduces classification at any fixed excess order to normal subgroups of one finite nilpotent quotient. The explicit square groups in the affine-square section provide a complementary construction: their orders vary, while their proper marked subgroups remain unchanged.

The companion applications use these mechanisms for prescribed two-power orders, three-generator hypertopes, square-section assemblies, common quotients, high-type classifications and the nondegenerate order-1024 census. Construction and presentation recognition are distinct: the former isolates local kernels in a compatible product, while the latter proves that a proposed presentation has no further kernel. Both steps are proved below. The source results retain their original attribution; no historical priority for these shared arguments is asserted.

2 Markings, images and kernels

A marked group is a group G together with a finite indexed generating family $(r_i)_{i\in I}$. Except where stated, the markers need not be involutions or distinct. For $S \subseteq I$, put

$$G_S = \langle r_i : i \in S \rangle, \qquad G_\varnothing = 1.$$

We call these subgroups parabolics, retaining their index sets even when two of them happen to coincide. The marking has the intersection property, or IP, if

$$G_S \cap G_T = G_{S\cap T} \quad (S, T \subseteq I). \tag{M1}$$

A marked homomorphism sends each source marker to the corresponding target marker. Its image on G_S is therefore the target parabolic with index set S.

All subgroup products below are products of sets unless a subgroup assertion is made. We write $\gamma_1(G) = G$ and $\gamma_{j+1}(G) = [\gamma_j(G), G]$. We use the elementary correspondence and isomorphism theorems for groups. In the finite p-group arguments we retain Burnside's basis theorem, the nilpotence of finite p-groups, and the usual multilinear commutator description of the lower-central graded Lie ring. Precisely, its degree-one part generates the graded Lie ring, and degree j is generated by left-normed commutators of weight j in any generating set. In the amalgam section we explicitly retain the normal-form theorem for an amalgamated free product. None of these classical inputs is presented as a result originating in this manuscript.

Lemma UA.K0 (lifting one intersection). Let $q : H \to Q$ be a homomorphism and let $C \le A\cap B$, where $A, B, C \le H$. If

$$q(A) \cap q(B) = q(C),$$

then

$$A \cap B = C \quad \Longleftrightarrow \quad A \cap B \cap \ker q \le C. \tag{M2}$$

Proof. The forward implication is immediate. For the converse, let $x \in A \cap B$. Its image belongs to $q(C)$, so choose $c \in C$ with $q(c) = q(x)$. Then $xc^{-1} \in A \cap B \cap \ker q$, which is contained in C by hypothesis. Multiplying by c gives $x \in C$. The inclusion $C \le A \cap B$ gives equality. □

Theorem UA.K (kernel profiles). Let $q : H \twoheadrightarrow Q$ be a marked epimorphism and suppose that the marking of Q has IP. Write $K = \ker q$ and $K_S = K \cap H_S$. Then H has IP if and only if

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I). \tag{M3}$$

Proof. Apply UA.K0 to $A = H_S$, $B = H_T$, and $C = H_{S\cap T}$. The image hypothesis is exactly IP in Q. The subgroup on the left of the containment in (M2) is $K_S \cap K_T$. Its containment in $H_{S\cap T}$ is equivalent to its containment in $K_{S\cap T}$; the reverse inclusion always holds. This proves each required intersection and its converse. □

The next result concerns the opposite direction. Passing to a quotient can create new intersections, and this defect is a product condition in its kernel.

Theorem UA.Q (exact intersections of images). Let $q : H \to Q$ be a homomorphism with kernel N, and let $A, B \le H$. Then

$$q(A) \cap q(B) = q(A \cap B) \quad \Longleftrightarrow \quad N \cap AB = (N \cap A)(N \cap B). \tag{M4}$$

If q is injective on A and B, the right side reduces to $N \cap AB = \{1\}$. If $N \le A$, the equality always holds.

Proof. Suppose the image intersection is exact, and take $ab \in N$, where $a \in A$ and $b \in B$. Since $q(a) = q(b)^{-1}$, choose $c \in A \cap B$ with this image. Then

$$ab = (ac^{-1})(cb) \in (N \cap A)(N \cap B).$$

The reverse inclusion requires no hypothesis. Conversely, if $q(a) = q(b)$, then $ab^{-1} \in N \cap AB$. Write $ab^{-1} = uv$, with $u \in N \cap A$ and $v \in N \cap B$. The element $u^{-1}a = vb$ lies in $A \cap B$ and has image $q(a)$, proving the required image equality. Under injectivity both factors on the right of (M4) are trivial. If $N \le A$, both sides of that equality are N. □

3 Deletion maps and supported extensions

A deletion for $S \subseteq I$ is an endomorphism $e_S : G \to G$ satisfying $e_S(r_i) = r_i$ for $i \in S$ and $e_S(r_i) = 1$ otherwise. Requiring these values matters: a general retraction onto G_S need not be a deletion.

Proposition UA.D (Boolean deletions). If a marked group has a deletion e_S for every $S \subseteq I$, then

$$e_S e_T = e_{S\cap T}, \qquad \operatorname{im} e_S = \operatorname{Fix}(e_S) = G_S. \tag{D1}$$

The marking has IP, and

$$\ker e_S = \langle\langle r_i : i \notin S \rangle\rangle_G, \qquad G = (\ker e_S) \rtimes G_S. \tag{D2}$$

Proof. The two maps in the first equality agree on every generator. The image is generated by precisely the retained markers, and each of those markers is fixed; this proves the image and fixed-point statements. An element of $G_S \cap G_T$ is fixed by $e_S e_T = e_{S\cap T}$, so belongs to $G_{S\cap T}$. The reverse inclusion is immediate.

Let M be the normal closure in (D2). Clearly $M \leq \ker e_S$. In G/M, every omitted marker is trivial, so the image of any word equals the image of its deletion. A word in $\ker e_S$ is therefore trivial in G/M, proving equality. Finally, for $g \in G$, the factorization $g = (g e_S(g)^{-1}) e_S(g)$ has factors in $\ker e_S$ and G_S. Their intersection is trivial because e_S fixes G_S. This is the asserted semidirect product. □

We now assemble extensions whose extra kernel is invisible on each proper part of its support. The groups and kernels may be infinite and nonabelian.

Theorem UA.SK (supported-kernel realization). Let I be finite and let $\mathcal{F}$ be a finite family of distinct subsets of I. Let $B = \langle b_i : i \in I \rangle$ have IP. For every $S \in \mathcal{F}$, suppose the following data are given:

1. A deletion homomorphism $p_S : B \to B_S$, fixing b_i for $i \in S$ and killing b_i otherwise.
2. A group $E_S = \langle e_{S,i} : i \in S \rangle$ and a marked epimorphism $q_S : E_S \to B_S$.
3. Injectivity of q_S on $(E_S)_T = \langle e_{S,i} : i \in T \rangle$ for every $T \subsetneq S$.

Put $N_S = \ker q_S$, and form the group

$$H = \left\{ (b, (x_S)_{S\in\mathcal{F}}) \in B \times \prod_{S\in\mathcal{F}} E_S : q_S(x_S) = p_S(b) \text{ for every } S \right\}. \tag{K1}$$

For $i \in I$, define h_i to have base coordinate b_i and S-coordinate $e_{S,i}$ when $i \in S$, and 1 otherwise. These elements generate all of H. If $\pi : H \to B$ is the base projection and $H_L = \langle h_i : i \in L \rangle$, then, for every $L \subseteq I$,

$$\pi(H_L) = B_L, \qquad \ker \pi = \prod_{S\in\mathcal{F}} N_S, \qquad \ker \pi \cap H_L = \prod_{\substack{S\in\mathcal{F} \\ S\subseteq L}} N_S. \tag{K2}$$

The factors on the right are embedded as pure coordinate subgroups. The marking of H has IP. If B and all E_S are finite, then

$$|H_L| = |B_L| \prod_{\substack{S\in\mathcal{F} \\ S\subseteq L}} |N_S|. \tag{K3}$$

Proof. The compatibility equations in (K1) are preserved by products and inverses, so H is a subgroup of the displayed direct product. The prescribed tuple h_i satisfies those equations by the literal deletion rule. Let H' be the subgroup generated by these tuples. Its base image is B, and the projection of $H'_S = \langle h_i : i \in S \rangle$ onto E_S is onto.

We first prove, by induction on inclusion among the finitely many supports, that every pure coordinate element of N_S belongs to H'_S. Fix $n \in N_S$. Choose a word on the markers indexed by S which

evaluates to n in E_S, and evaluate the same word in H', obtaining $w \in H'_S$. Its base coordinate lies in B_S and is $q_S(n) = 1$.

Consider its coordinate at another support T. Because the word uses only markers from S, that coordinate lies in $(E_T)_{S\cap T}$. Its image under q_T is $p_T(1) = 1$. If $T \not\subseteq S$, then $S \cap T \subsetneq T$, and the proper-parabolic injectivity hypothesis forces this coordinate to be 1. The only possible nonidentity coordinates of w are consequently at supports $T \subseteq S$. Its S-coordinate is the prescribed n, while every coordinate at a proper support $T \subsetneq S$ belongs to N_T.

For each such proper support, the induction hypothesis supplies the pure coordinate element having exactly that value in N_T, as an element of $H'_T \leq H'_S$. Multiply w by the inverses of those elements. Distinct pure coordinate subgroups commute in the ambient direct product, so these operations remove the unwanted coordinates without changing the S-coordinate. The result is the pure element n in coordinate S, and remains in H'_S. A minimal support has no smaller coordinates to remove, so this argument includes the initial step. No commutativity within an individual kernel has been used.

The kernel of π in the full group H is exactly the direct product of the N_S, by (K1). The induction places that entire kernel in H'. Since H' already maps onto B, every $h \in H$ has the same base coordinate as some $h' \in H'$; then $h(h')^{-1} \in \ker\pi \leq H'$. Thus $H' = H$.

The image equality $\pi(H_L) = B_L$ follows on generators. If $x \in H_L \cap \ker\pi$, its S-coordinate lies in $(E_S)_{L\cap S}$ and in N_S. This coordinate is trivial whenever $S \not\subseteq L$, again by proper-parabolic injectivity. Conversely, when $S \subseteq L$, the isolation argument placed the entire pure N_S in $H_S \leq H_L$. This proves all three assertions in (K2).

The coordinate products in (K2) satisfy

$$\left(\prod_{S\subseteq L} N_S\right) \cap \left(\prod_{S\subseteq M} N_S\right) = \prod_{S\subseteq L\cap M} N_S.$$

Theorem UA.K now proves IP in H. More explicitly, an element of $H_L \cap H_M$ has base in $B_{L\cap M}$; subtracting a lift from $H_{L\cap M}$ leaves an element in both displayed kernel products, hence in the kernel product for $L \cap M$. This also shows exactly where IP of the base is used. Finally (K3) is the order formula for the surjection $H_L \to B_L$ with the kernel just computed. □

Corollary UA.SK.Local (recognition on an antichain). Under UA.SK, if $\mathcal{F}$ is an antichain under inclusion, then the coordinate projection $H_S \to E_S$ is a marked isomorphism for every $S \in \mathcal{F}$.

Proof. It is onto on generators. If an element maps to 1, its base coordinate also vanishes, since the base value of a word on S is the q_S-image of its S-coordinate. Formula (K2) says that the remaining kernel in H_S has only the factor N_S: the antichain excludes every other support contained in S. The S-coordinate detects that factor, so the element is 1. □

Corollary UA.SK.Delete (derived deletions). If, in addition, B has deletions for every subset of I, then every E_S and H has all its Boolean deletions.

Proof. For $T \subsetneq S$, the restriction $q_S : (E_S)_T \to B_T$ is a marked isomorphism: it is onto on generators and injective by hypothesis. Compose q_S, the deletion $B_S \to B_T$, and the inverse of this isomorphism. This gives a homomorphism $E_S \to (E_S)_T$ that fixes exactly the T-markers and kills the others. For $T = S$ use the identity. Proposition UA.D supplies the Boolean identities.

For $L \subseteq I$, apply the base deletion e_L and, in coordinate S, the local deletion onto $S \cap L$. The two sides of each compatibility equation agree on the local markers; the base deletions commute by UA.D. Thus the coordinatewise operation preserves (K1). On h_i it fixes h_i if $i \in L$, and kills it otherwise. It is the required deletion of H. □

Corollary UA.SK.Central (lower-central kernel). Under UA.SK, suppose $c \geq 1$, $\gamma_c(B) = 1$, and $N_S = \gamma_c(E_S)$ for every support. Then

$$\gamma_c(H) = \ker\pi. \tag{K4}$$

Proof. Applying π to a commutator of weight c gives 1, so $\gamma_c(H) \leq \ker\pi$. For the converse, repeat the inclusion induction in the proof of UA.SK. The projection $H_S \to E_S$ is onto; therefore its image on $\gamma_c(H_S)$ is $\gamma_c(E_S) = N_S$. Given $n \in N_S$, choose a lift $w \in \gamma_c(H_S)$. Its base is 1, since $\gamma_c(B) = 1$. The

same proper-support argument makes all coordinates at $T \not\subseteq S$ trivial. The remaining proper-support coordinates lie in N_T; by induction their pure copies belong to $\gamma_c(H)$. Removing them leaves the pure n, still in $\gamma_c(H)$. All pure kernels are obtained, proving (K4). □

Corollary UA.SK.Separator (factorization across a separator). Under UA.SK, let $I = L \cup M$. Suppose $B = B_L B_M$ as a product of sets, and every support in $\mathcal{F}$ is contained in L or in M. Then $H = H_L H_M$.

Proof. For $h \in H$, factor $\pi(h) = b_L b_M$ and choose lifts $u_L \in H_L$, $u_M \in H_M$. Then $h = n u_L u_M$ for some $n \in \ker \pi$. Assign every pure coordinate factor of n to one of the two sides containing its support. Formula (K2) gives $n = n_L n_M$, with $n_L \in H_L$, $n_M \in H_M$. Each pure N_S is normal in H, because conjugation in the ambient product acts coordinatewise and $N_S \trianglelefteq E_S$. Hence the product of the assigned M-factors is normal in H. Moving n_M past u_L gives

$$h = (n_L u_L)(u_L^{-1} n_M u_L \, u_M) \in H_L H_M.$$

The opposite inclusion is automatic. □

The hypotheses of UA.SK do not identify H with a universal amalgam of the E_S. Additional cross-relations and a presentation-recognition argument may be necessary. Nor can injectivity on every proper parabolic be replaced by injectivity on individual markers alone.

Proposition UA.Fiber (two compatible split factors). Let A and B be marked groups on finite label sets L and M, and put $J = L \cap M$. Suppose their J-parabolics are identified with the same marked group C, and that $p_A : A \to C$ and $p_B : B \to C$ are deletions. Then

$$A \times_C B \cong (\ker p_A \times \ker p_B) \rtimes C. \tag{F1}$$

This full fiber product is generated by the tuples $(a_i, 1)$ for $i \in L \setminus J$, $(1, b_i)$ for $i \in M \setminus J$, and (c_i, c_i) for $i \in J$. The two original groups embed through $a \mapsto (a, p_A(a))$ and $b \mapsto (p_B(b), b)$. If finite, its order is $|A||B|/|C|$.

Proof. Every compatible pair has a unique form $(n_A c, n_B c)$, giving the split decomposition and the order formula. The stated copies of A and B are generated by subsets of the stated tuple marking. They contain $(n_A, 1)$ and $(1, n_B)$, respectively, while either contains the diagonal copy of C. These subgroups generate the split product, proving the assertion. The embedding formulas are injective because their first and second projections, respectively, are identities. □

4 Recognition of presentations and compatible products

All homomorphisms in this paper are group homomorphisms. A product of subgroups means a product of sets until it has been shown to be a subgroup. No group is assumed finite in this section.

Theorem UA.Normal (normal kernel recognition). Let $\pi : P \to B$ and $\rho_i : P \to E_i$, for i in a finite set J, be homomorphisms. Suppose $K_i \trianglelefteq P$ satisfy

$$\ker \pi = \langle K_i : i \in J \rangle, \tag{1}$$

$$\rho_i(K_j) = 1 \quad (i \neq j), \tag{2}$$

$$\ker \rho_i \cap K_i = 1. \tag{3}$$

Then the homomorphism

$$P \longrightarrow B \times \prod_{i \in J} E_i, \qquad x \longmapsto (\pi(x), (\rho_i(x))_i)$$

is injective. Moreover, the kernel in (1) is the internal direct product of the K_i. For injectivity alone it suffices to replace (1) by $\ker \pi \leq \langle K_i : i \in J \rangle$.

Proof. For $i \neq j$, an element of $K_i \cap K_j$ is killed by ρ_i by (2), and is therefore trivial by (3). Normality gives $[K_i, K_j] \leq K_i \cap K_j$, so distinct factors commute. Their finite set product is consequently their generated subgroup. If $\prod_i k_i = 1$ with $k_i \in K_i$, applying ρ_j gives $\rho_j(k_j) = 1$, whence $k_j = 1$. Thus multiplication identifies the external product with the generated subgroup. Finally, an element in the kernel of the displayed homomorphism belongs to $\ker \pi$, hence has such a product expression. Every coordinate vanishes, so every factor is trivial. □

Corollary UA.RecognizeQuotient. Let $f : P \twoheadrightarrow H$ be a homomorphism, and let H have a base homomorphism $p : H \to B$ and coordinates $c_i : H \to E_i$. If UA.Normal applies to pf, $c_i f$ and normal subgroups $K_i \leq P$, then f is an isomorphism.

Proof. The coordinate map factors through f, so injectivity of the coordinate map implies injectivity of f. Surjectivity is the stated hypothesis. □

Normality is substantial. Let $P = D_8 = \langle u, v \mid u^2 = v^2 = (uv)^4 = 1 \rangle$, let $B = 1$, and take $K_1 = \langle u \rangle$, $K_2 = \langle v \rangle$. Its abelianization $P \to C_2^2$ has coordinates faithfully detecting the two factors and killing the other one. The factors generate P, but the map has the nontrivial kernel $\langle (uv)^2 \rangle$. Here neither factor is normal. Thus a correct local model and coordinate detection do not themselves eliminate mixed relations.

The following specialization uses the full compatible product H in UA.SK and assumes that its support family is an antichain.

Theorem UA.Recognition (complete presentation). In this setting, let $P = \langle r_t : t \in I \rangle$ be given by a presentation. Suppose:

1. its relators hold on the marker tuples in the compatible product H;
2. the local relations of every E_S hold on $(r_t)_{t \in S}$;
3. the image K_S of N_S under $E_S \to P$ is normal in P;
4. adjoining the relations $K_S = 1$ for all S gives exactly the marked presentation of B.

Then the marker map $P \to H$ is an isomorphism. In particular the K_S form an internal direct product, all marked intersections are exact, and if the input groups are finite then

$$|P_L| = |B_L| \prod_{S \subseteq L} |N_S|. \tag{4}$$

The additional deletion conclusion of UA.SK.Delete transfers to P under its hypotheses. The same holds for the geometric conclusion of UA.SK.Geometry under its hypotheses.

Proof. The first assumption and the generation assertion of UA.SK give a marked surjection $f : P \twoheadrightarrow H$. The second gives $E_S \to P_S$; its composite with f and projection to E_S is the identity on every local marker, hence is the identity homomorphism. Thus each local map is injective, without an order comparison. Under f, the image of N_S is its pure S-coordinate: the antichain makes every other intersection of supports proper, where q_T is injective. Consequently the coordinate maps kill K_S outside its own coordinate and are injective on K_S in that coordinate.

Normality makes $K = \langle K_S : S \in \mathcal{S} \rangle$ a normal subgroup. The last assumption identifies the marked map $P/K \to B$ with an isomorphism, so the base map $P \to B$ has kernel exactly K. UA.RecognizeQuotient now proves that f is an isomorphism. The remaining conclusions follow from UA.SK and the explicitly indicated additional clauses. □

The last assumption is a presentation calculation. It does not assume the isomorphism $P \cong H$ which is being proved. In practice it says that, after killing the displayed local kernel words, the remaining relations are the known base presentation. A pair of marked maps which are inverse on generators

also proves this condition. Normality likewise has a concrete local test: the markers within S already normalize K_S, and one must check the remaining markers. It is enough that those remaining markers centralize a generating set of K_S. Merely writing down the desired quotient does not supply this check.

Proposition UA.Mix (jointly faithful deletions). Let $G = \langle r_i : i \in I \rangle$ have every Boolean deletion e_J. Suppose $I = L \cup M$, put $C = L \cap M$, and assume $\ker e_L \cap \ker e_M = 1$. Then

$$G \longrightarrow G_L \times_{G_C} G_M, \qquad g \longmapsto (e_L(g), e_M(g))$$

is a marked isomorphism. The maps to G_C on the right are the restrictions of the corresponding deletions. No group is assumed finite.

Proof. UA.D gives $e_L e_M = e_C = e_M e_L$. Hence the displayed pair is compatible, and the homomorphism's kernel is the stated trivial intersection. For a compatible pair $(x, y) \in G_L \times_{G_C} G_M$, put $c = e_C(x) = e_C(y)$. Since the two deletions fix their respective parabolics, $e_M(x) = c = e_L(y)$ and both deletions fix c. Thus

$$e_L(xc^{-1}y) = xc^{-1}c = x, \qquad e_M(xc^{-1}y) = cc^{-1}y = y.$$

Every compatible pair therefore has the explicit preimage $xc^{-1}y$. The homomorphism is surjective as well as injective, with inverse given by this formula. Its values on the markers give the stated marking. □

The kernel condition can be checked by coordinates. In a supported-kernel realization with inherited deletions, suppose every base coordinate is detected on at least one side, and every support is contained in L or M. An element killed by both deletions has trivial base coordinate. Each pure supported-kernel coordinate is then detected on a side containing its support, so all these coordinates are trivial too. This proves the kernel condition in the path and two-branch applications below; it is not assumed from a coincidence of finite orders.

5 Coordinate kernels and incidence geometries

An incidence system has a finite type set, objects carrying types, and a symmetric incidence relation; distinct objects of one type are not incident. A flag is a set of pairwise incident objects. A chamber has one object of each type. A geometry is an incidence system in which every flag extends to a chamber. The residue of a flag consists of the objects of the remaining types incident with that flag, with induced incidence. The geometry is residually connected when the incidence graph of every residue of rank at least two, including the empty-flag residue, is connected. It is thin when every panel, a flag missing just one type, lies in exactly two chambers. A regular hypertope is a thin, residually connected geometry admitting a chamber-transitive type-preserving automorphism group.

For a marked group indexed by I, its maximal parabolics are $P_i = G_{I\setminus\{i\}}$. The associated typed coset incidence system has objects $P_i g$, and incidence means nonempty intersection of cosets. All actions and cosets below use this right-coset convention.

For subgroups $P_i \leq G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Proposition UA.Geometry (coset recognition and faithful chamber action). Suppose the marking of G consists of nonidentity involutions, has IP, and has typed coset Helly for its maximal parabolics. The coset incidence system is then a thin, residually connected geometry, and G is its full type-preserving automorphism group, acting regularly on chambers. More generally, the full type-preserving automorphism group of any thin, residually connected geometry of finite rank acts freely on chambers. A chamber-transitive subgroup is consequently the full group.

Proof. In the coset system, a common coset point extends every flag to a chamber by choosing the coset of each missing type through that point. Right translation sends every such flag to its base flag. For a base flag of type set S, IP gives stabilizer $G_{I\setminus S}$. In particular the full chamber stabilizer is 1, while a panel stabilizer is a group generated by one nonidentity involution and has order two. There are therefore exactly two chambers through each panel.

Consider a residue with remaining type set T, where $|T| \geq 2$. Its stabilizer is G_T, and its maximal object stabilizers are $G_{T\setminus\{i\}}$, $i \in T$. Every marker indexed by T lies in at least one of these proper parabolics. A word in those subgroups supplies a walk between their translated base objects in the residue incidence graph; all objects are translates of base objects and all base objects lie in a common flag. The residue is therefore connected.

For the general assertion, first observe that every residually connected finite-rank geometry has a connected chamber graph. This follows by induction on rank: chambers through a fixed object are connected through galleries in its residue, and a path in the incidence graph, extending each consecutive incident pair to a chamber, connects arbitrary chambers through these subgraphs. The rank-one case holds because all chambers share the empty panel. In a thin geometry an automorphism fixing a chamber fixes its unique neighbor across each panel. It therefore fixes every chamber along every gallery, and hence every object. The chamber action is free. If a subgroup is chamber-transitive, any full automorphism agrees on a base chamber with one subgroup element; their quotient fixes that chamber and is the identity. Apply this to the coset geometry to obtain the stated full automorphism group. □

5.1 Helly transfer through coordinate kernels

Write right cosets as Pg, with the subgroup on the left. Let $q : H \to B$ be an epimorphism of groups generated by corresponding marked families $(h_i)_{i\in I}$ and $(b_i)_{i\in I}$, where I is finite. For $J \subseteq I$, write $H_J = \langle h_j : j \in J\rangle$ and $B_J = \langle b_j : j \in J\rangle$. In particular, $q(H_J) = B_J$ for every J.

Fix any set $\mathcal{J}$ of subsets of I as the permitted types. The maximal-parabolic case is $\mathcal{J} = \{I \setminus \{i\} : i \in I\}$.

Theorem UA.SK.Helly (coordinate-kernel transfer). Suppose B has IP. Suppose $N = \ker q$ is an internal direct product $\prod_{\alpha\in A} N_\alpha$, with a finite factor set A, and for every $J \in \mathcal{J}$,

$$N \cap H_J = \prod_{\alpha\in A_J} N_\alpha \qquad \text{for some } A_J \subseteq A. \tag{H1}$$

If the family $(B_J)_{J\in\mathcal{J}}$ has typed coset Helly, then so does $(H_J)_{J\in\mathcal{J}}$.

No commutativity within the factors, finiteness of H or B, or faithfulness of an action of B on N is assumed. The required intersection-image fact is $q(H_{J\cap K}) = B_J \cap B_K$; it follows here from the marked map and IP of B. IP of H is not an additional assumption for this transfer theorem.

Proof. Consider pairwise intersecting cosets $H_{J_i}g_i$. Their images $B_{J_i}q(g_i)$ intersect pairwise. Choose a common base element b by the assumed Helly property, and choose u in H with $q(u) = b$. Each set

$$F_i = H_{J_i}g_i \cap q^{-1}(b)$$

is nonempty: because $bq(g_i)^{-1} \in B_{J_i}$, a preimage of that element in H_{J_i}, multiplied on the right by g_i, belongs to F_i.

We first show that $F_i \cap F_j \neq \emptyset$. Choose $x \in H_{J_i}g_i \cap H_{J_j}g_j$ and put z=q(x). Both b and z belong to each of the two base cosets, so

$$bz^{-1} \in B_{J_i} \cap B_{J_j} = B_{J_i\cap J_j}.$$

The marked map is surjective on this intersection parabolic. Choose $h \in H_{J_i\cap J_j}$ with $q(h) = bz^{-1}$. Since h belongs to both H_{J_i} and H_{J_j}, left multiplication by h preserves both right cosets. Thus hx belongs to their intersection and has image b. This proves the assertion without assuming that q carries an arbitrary intersection onto the intersection of its images.

Choose $x_i \in F_i$ and put $k_i = x_i u^{-1} \in N$. If $y \in F_i$, then $yx_i^{-1} \in N \cap H_{J_i}$; the converse also holds. Hence

$$F_i u^{-1} = (N \cap H_{J_i}) k_i. \tag{H2}$$

Under the direct product coordinates of N, (H1) makes (H2) unrestricted in coordinates $\alpha \in A_{J_i}$, and fixes coordinate $\alpha \notin A_{J_i}$ to $(k_i)_\alpha$. Pairwise intersections imply that any two constraints fixing the same coordinate fix it to the same value. Choose that value in every constrained coordinate and the identity in every unconstrained coordinate. The resulting element belongs to all sets in (H2). Multiplication on the right by u supplies a common element of the original cosets. This proves the theorem. □

Corollary UA.SK.Geometry. In the supported-kernel realization UA.SK, assume the base marking gives a regular hypertope and every marked generator in every attached group is an involution. Then the resulting marking also gives a regular hypertope, and its full type-preserving automorphism group is the constructed group.

Proof. UA.SK gives IP and the exact profile

$$N \cap H_J = \prod_{S \subseteq J} N_S.$$

The base maximal-parabolic cosets have typed Helly. Theorem UA.SK.Helly applies with factors indexed by the supports S. Each constructed generator is a tuple whose base coordinate and every nontrivial attached coordinate are involutions; hence its square is the identity. Its base image is nonidentity, so its order is exactly two. UA.Geometry now gives thinness, residual connectedness, chamber regularity and equality with the full type-preserving automorphism group. In particular, this conclusion uses typed Helly in addition to IP. □

A sufficient alternative to local involutivity is that every support has size at least two: each singleton local parabolic is then proper, and injectivity into its nonidentity involutory base image forces the local marker to have order two. This is a sufficient condition, not an equivalent reformulation. Without either condition the geometric corollary fails. With a singleton type set and support, the quotient $C_4 \to C_2$ is injective on its only proper parabolic, the trivial subgroup. UA.SK then constructs C_4, whose distinguished generator has order four. In rank three, adjoining two independent C_2 markers gives $C_4 \times C_2^2$ with the same defective singleton attachment and a four-element panel.

6 Frattini coordinates

Theorem UA.Phi (primitive saturation and coordinates). Let P be a finite p-group with an irredundant generating marking $(r_i)_{i \in I}$, $|I| = d$, and let $F = \Phi(P)$. Then, for every $S \subseteq I$,

$$\Phi(P_S) = P_S \cap F, \qquad P_S F = \pi^{-1} \operatorname{span}(\bar{r}_i : i \in S), \qquad [P : P_S F] = p^{d-|S|}. \tag{P1}$$

The saturated parabolics $P_S F$ intersect according to intersection of index sets. Each omission subgroup $P_{I \setminus \{i\}}$ has exactly one maximal overgroup, namely $P_{I \setminus \{i\}} F$.

If $B \geq F$ and $(s_i)_{i \in J}$ is an irredundant relative generating family over B, then $B \trianglelefteq P$, P/B has basis $(s_i B)_{i \in J}$ over $\mathbb{F}_p$, and

$$\langle B, s_i : i \in T \rangle / B = \operatorname{span}(s_i B : i \in T). \tag{P2}$$

If $p = 2$ and all original r_i are involutions, then $F = P'$ and $\Phi(P_S) = P_S'$ for every S.

Proof. Burnside's basis theorem makes the $\bar{r}_i$ a basis of P/F. The image of P_S therefore has order $p^{|S|}$, with kernel $P_S \cap F$. Since this image is elementary abelian, $\Phi(P_S) \leq P_S \cap F$. On the other hand P_S is generated by $|S|$ elements, so Burnside gives $[P_S : \Phi(P_S)] \leq p^{|S|}$. The quotient by $P_S \cap F$ already has that order. Equality follows, proving the first formula. The remaining formulas and the intersection assertion are inverse-image statements for coordinate subspaces.

Every maximal subgroup of a finite p-group contains F and corresponds to a hyperplane of P/F. A hyperplane containing a specified coordinate hyperplane equals it, proving uniqueness of the maximal overgroup of an omission subgroup.

For the relative assertion, every subgroup containing F is normal, because P/F is abelian. The quotient P/B is elementary abelian. The indicated images span it, and any nontrivial linear relation would express one s_i using B and the other markers, contradicting irredundancy. They form a basis, proving (P2). Finally, an abelian group generated by involutions is elementary abelian of exponent two. Thus $P' \le \Phi(P)$, while the elementary abelian quotient P/P' gives the reverse inclusion. The same argument applies to each P_S. □

These coordinates also describe the corresponding omission cosets: a coset prescribes one coordinate value, and every simultaneous assignment gives a chamber. The geometry is complete multipartite, with p objects in each type, and the quotient translation group acts regularly on chambers. For $p > 2$ it need not be the full type-preserving automorphism group; the thin case $p = 2$ is the one covered by UA.Geometry.

7 The canonical quotient of a forest marking

Suppose a group is generated by nonidentity involutions $r_i, i \in I$. Its noncommutation graph has vertex set I, and an edge $\{i, j\}$ precisely when r_i and r_j do not commute. We write $G_J = \langle r_j : j \in J \rangle$, as in the marking definitions. The intersection property in this section always refers to the original group. It is not assumed to survive an intermediate quotient.

Lemma UA.Lift (descending lift). Let G be a finite nilpotent group, let $A, B \le G$ satisfy $A \cap B = 1$, and fix an integer $m \ge 1$. Put $D_j = \gamma_j(G), A_j = A \cap D_j, B_j = B \cap D_j$. If the images of A_j, B_j generate D_j/D_{j+1} for every $j \ge m$, then $D_j = A_j B_j$ for every $j \ge m$.

Proof. Descend from a trivial term. Assuming $D_{j+1} = A_{j+1}B_{j+1}$, if $a \in A_j, b \in B_j$ have equal images in the next quotient, write $ab^{-1} = a'b'$ with $a' \in A_{j+1}, b' \in B_{j+1}$. Then $a'^{-1}a = b'b \in A \cap B$, so both images are trivial. Their images therefore form an internal direct sum in the abelian quotient. Counting gives $|D_j| = |A_j||B_j| = |A_jB_j|$, proving equality as sets. No prior normality or permutability of A, B is assumed. ■

Lemma UA.Support (graded support). Let a positively graded Lie algebra be generated in degree one by disjoint sets X, Y, with subalgebras L_X, L_Y. If $[x, y] \in L_Y$ for $x \in X, y \in Y$, then $L = L_X + L_Y$, degree by degree.

Proof. The derivation $\operatorname{ad} x$ preserves L_Y because it does so on its generators. Iterated brackets of those derivations also preserve L_Y, by Jacobi. Hence $[L_X, L_Y] \subseteq L_Y$, and $L_X + L_Y$ is a Lie subalgebra containing all generators. The argument holds in characteristic two. ■

Theorem UA.Forest (forest quotient). Suppose G is a finite two-group generated by nonidentity involutions r_i, with marked IP, and its noncommuting pairs form a forest T on d vertices and m edges. Then edge commutators are independent in $G'/\gamma_3(G)$, and $G/\gamma_3(G)$ is the marked group

$$(v, z)(w, t) = (v + w, z + t + \beta(v, w)), \quad v, w \in \mathbb{F}_2^d,\ z, t \in \mathbb{F}_2^m, \tag{F0}$$

where, after orienting each edge, its component of β is $v_{\text{head}} w_{\text{tail}}$. Its order is 2^{d+m}.

Proof. IP gives irredundancy. UA.Phi gives $G/G' = C_2^d$ and $G' \cap \langle r_i \rangle = 1$. The lower-central graded Lie ring is generated by these degree-one elements; all its graded components are $\mathbb{F}_2$-spaces by bilinearity. Edge commutators span degree two. If they satisfy a nonzero linear relation, root each forest component and choose a participating edge whose child endpoint has maximal depth in its component. Removing it separates the descendant set X from all other vertices Y, and no other participating edge is internal to X. The relation therefore writes the only nonzero cross bracket as a sum of brackets internal to Y. Lemma UA.Support shows every graded component is generated by pure X- and pure Y-brackets. Their representatives lie in $G_X \cap \gamma_j(G)$ and $G_Y \cap \gamma_j(G)$.

Actual IP gives $G_X \cap G_Y = 1$, so Lemma UA.Lift gives $G' = (G' \cap G_X)(G' \cap G_Y)$. For the crossing endpoints $x \in X, y \in Y$, write $[r_x, r_y] = uv$ accordingly. Then $u \in G_X \cap G_{\{x\} \cup Y} = \langle r_x \rangle$, whence $u = 1$; next $[r_x, r_y] \in G_Y \cap G_{\{x,y\}} = \langle r_y \rangle$, so it too is one. This contradicts that the edge is noncommuting. Independence follows.

Bilinearity makes (F0) associative; its coordinate markers are involutions with independent central edge commutators. It is generated by those markers, and its parabolic on any subset J consists exactly of vertex coordinates supported on J and edge coordinates supported on edges internal to J. Thus it has IP. Restricting the vertex coordinates to a subset and retaining exactly its internal edge coordinates preserves the multiplication formula, so it also gives every Boolean deletion. Collection in the corresponding class-two presentation gives at most 2^{d+m} elements, and this coordinate model attains the bound. Independence in $G'/\gamma_3(G)$ gives the same order for $G/\gamma_3(G)$; hence the marked map is an isomorphism. ■

8 Finite quotients of prescribed excess order

The next theorem separates a universal finite search space from the task of proving that a particular family has the required class-two quotient. That identification is an application hypothesis, not a consequence of the search.

Theorem UA.Nilpotent (complete marked kernel parameter space). Let $F = \langle x_i : i \in I \rangle$, where $|I| = d$ and $x_i^2 = 1$. Suppose F has all Boolean deletions. Put $Q = F/\gamma_3(F)$, with its image marking. For $k \geq 0$, set

$$U = F/\gamma_{k+3}(F), \qquad K = \gamma_3(U), \qquad K_S = K \cap U_S. \tag{N1}$$

Then U is a finite two-group,

$$|U| \leq 2^{\sum_{j=1}^{k+2} d^j}, \tag{N2}$$

the markings of U and Q have Boolean deletions and IP, and U/K is canonically the marked group Q. For every $S \subseteq I$,

$$K_S = \gamma_3(U_S). \tag{N3}$$

There is a bijection between the following two sets:

1. Ordered marked isomorphism classes of finite two-group quotients $f : F \twoheadrightarrow G$ for which the induced marked map $Q \to G/\gamma_3(G)$ is an isomorphism, $|G| = |Q|2^k$, and G has IP.
2. Normal subgroups $N \trianglelefteq U$ satisfying $N \leq K$, $[K : N] = 2^k$, and

$$\frac{NK_S}{N} \cap \frac{NK_T}{N} = \frac{NK_{S\cap T}}{N} \quad (S, T \subseteq I). \tag{N4}$$

The correspondence sends N to U/N with the original image markers. If $D \leq \mathrm{Sym}(I)$ is specified and each $\sigma \in D$ induces an automorphism $x_i \mapsto x_{\sigma(i)}$ of F, then equivalence allowing precisely these label permutations is the D-orbit relation on the kernels in item 2.

Proof. The degree-one part of the lower-central graded Lie ring of F is generated by elements of exponent dividing two. By multilinearity, every iterated commutator in those generators also has exponent dividing two in its homogeneous component. In degree j, at most d^j left-normed commutators suffice to generate that component. Consequently

$$|\gamma_j(F)/\gamma_{j+1}(F)| \leq 2^{d^j}.$$

Multiplying these bounds for $1 \leq j \leq k+2$ proves finiteness, the two-power order and (N2). Each deletion preserves every lower-central term, and therefore descends both to U and to Q. Proposition UA.D gives IP for both image markings; no separate IP hypothesis on Q is needed. The epimorphic-image property of the lower central series gives $U/K \cong F/\gamma_3(F) = Q$.

For (N3), one inclusion follows from $U_S \leq U$. If $z \in \gamma_3(U) \cap U_S$, the deletion e_S fixes z and sends $\gamma_3(U)$ into $\gamma_3(U_S)$. This proves the reverse inclusion.

Let G satisfy item 1. Its subgroup $\gamma_3(G)$ has order 2^k. In a nilpotent group, equality between two successive nontrivial lower-central terms would persist at every later term, contradicting termination of that series. Thus each nontrivial step beginning at $\gamma_3(G)$ strictly decreases its order. There are at most k such steps, and $\gamma_{k+3}(G) = 1$. The map $F \to G$ therefore factors through U. Its kernel N lies in K, since the induced map $U/K \to G/\gamma_3(G)$ is an isomorphism, and the orders give $[K : N] = 2^k$.

Conversely, let $N \trianglelefteq U$ with $N \leq K$ and $[K : N] = 2^k$, and write $G = U/N$. Its third lower-central subgroup is K/N, its canonical class-two quotient is Q, and its order is $|Q|2^k$. If $q : U \to G$, then

$$(K/N) \cap G_S = q(K \cap U_S) = NK_S/N. \tag{N5}$$

Indeed, $q(u) \in K/N$ with $u \in U_S$ implies $u \in K$, because $q^{-1}(K/N) = K$. The reverse inclusion is immediate. Applying UA.K to $G \to Q$ proves that IP in G is equivalent to precisely (N4).

An isomorphism preserving each ordered marker commutes with the two maps from U, because it agrees with them on a generating family. Their kernels are consequently equal. Equal kernels plainly give the same marked quotient. Finally, every specified label permutation acts on F, preserves its characteristic lower-central terms, and hence acts on U and on the displayed kernels. A marker-permuting isomorphism induces the corresponding kernel equality after this automorphism, and the converse descends that automorphism to the quotients. This proves the orbit assertion at exactly the specified equivalence scope. □

Theorem UA.Ladder (invariant index-two descent). Let U be a finite two-group and $M \trianglelefteq U$. The U-normal subgroups $L \leq M$ of index two are exactly the inverse images of the hyperplanes of

$$M/\big(\Phi(M)[M, U]\big). \tag{N6}$$

For $K \trianglelefteq U$, every U-normal subgroup $N \leq K$ of index 2^k is obtained by k successive such steps.

Proof. An index-two subgroup L of M contains $\Phi(M)$ and has quotient C_2. If L is U-normal, conjugation by U on M/L is trivial, since $\mathrm{Aut}(C_2) = 1$. Thus $[M, U] \leq L$. Conversely, a hyperplane inverse image has index two and contains $[M, U]$; for $l \in L$ and $u \in U$, the conjugate of l differs from l by an element of $[M, U]$, so lies in L. The quotient in (N6) is elementary abelian by Burnside's basis theorem, proving the first assertion.

Suppose $N \trianglelefteq U$ and $N < M$. The nontrivial finite two-group M/N has a nonzero Frattini quotient V. Conjugation by U/N acts on the nonzero linear maps $V \to \mathbb{F}_2$. If $\dim V = a$, there are $2^a - 1$ such maps, an odd number. Every orbit has two-power size, so at least one orbit has size one. The kernel of a fixed nonzero map lifts to a U-normal subgroup between N and M, of index two in M. Repeating with the new subgroup gives a chain terminating at N; its length is k, since the total index is 2^k. □

Lemma UA.Order (presentation recognition by a sharp bound). Let $A_1, \dots, A_t$ be finite normal subgroups of G, with $|A_j| \leq a_j$. Suppose $[G : A_1 \cdots A_t] \leq b$. Then $|G| \leq b \prod_j a_j$. If a finite quotient of G has this latter order, its quotient map is an isomorphism.

Proof. Normality makes each partial product a subgroup. The multiplication map from $A_1 \times \cdots \times A_t$ onto the product is surjective as a map of sets, so that product has order at most $\prod_j a_j$. Multiplication by the quotient index proves the bound and finiteness of G. A quotient attaining the bound forces equality of the finite orders and hence a trivial kernel. □

Corollary UA.Search (rank and excess classify the finite search). Let T be a specified forest on d vertices and m edges, and let k≥0. Set

$$F_T = \langle x_i \mid x_i^2 = 1,\ [x_i, x_j] = 1 \text{ for nonedges} \rangle, \quad U_{T,k} = F_T/\gamma_{k+3}(F_T), \quad K = \gamma_3(U_{T,k}).$$

The ordered IP markings by nonidentity involutions with graph T and order 2^{d+m+k} are exactly the quotients $U_{T,k}/N$, where N is normal, contained in K, $[K : N] = 2^k$, and the images $(N(K \cap (U_{T,k})_J))/N$ have IP for every pair J,L. Generator-set equivalence is exactly the action of Aut(T) on these kernels.

Proof. Killing any set of generators preserves every defining relator of F_T. Collection in its class-two quotient gives at most d+m binary coordinates, and (F0) realizes all of them, so this quotient is precisely the marked model Q_T. Every target has that same quotient by UA.Forest. These are exactly UA.Nilpotent's hypotheses. Its all-parabolic condition and UA.Ladder give the stated bijection and exhaustive search. An ordered isomorphism must commute with the generating map, hence has the same kernel. A permutation of the generator set preserves the noncommutation graph and extends to an automorphism of F_T and $U_{T,k}$; this proves the last claim. ■

The last corollary is a finite classification procedure, with the marking retained. First form the finite group $U_{T,k}$; then iterate UA.Ladder inside $\gamma_3(U_{T,k})$, test the explicit intersection condition of UA.Nilpotent, and take the declared graph-automorphism orbits. The theorem proves that this procedure terminates and misses no target marking. Any reported numbers of groups, named presentations or universal orders still require execution and an application-specific identification. No numerical census is assumed or reported in this paper.

9 Affine square groups

Let

$$a(x,y) = (-x,y), \quad b(x,y) = (y,x), \quad c(x,y) = (x,1-y).$$

Proposition UA.Affine (the complete square presentation). These generate $W = \mathbb{Z}^2 \rtimes D_8$, of type $\{4,4\}$, with $ac = ca$. Its presentation is

$$W = \langle a,b,c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1\rangle. \tag{S1}$$

Proof. Put $d = bab$, $y_0 = cd$, and $x_0 = by_0b$. The relations give $[a,d] = 1$, $[a,y_0] = 1$, $dy_0d = y_0^{-1}$, $ax_0a = x_0^{-1}$, and conjugation by b swaps x_0, y_0. Substitution of $bc = x_0ab$ makes $(bc)^4 = 1$ equivalent to $[x_0,y_0] = 1$. Thus every word collects to $x_0^u y_0^v M$, $M \in \langle a,b\rangle = D_8$. In the affine model x_0, y_0 are the unit translations, so these forms are distinct. This proves (S1). □

The normal form also fixes our affine multiplication convention: the rightmost transformation acts first. The symbol D_8 denotes the group of signed permutation matrices in two coordinates, of order eight. In all dihedral notation below the subscript denotes the order of the group.

Proposition UA.Square (finite square seeds). For $n \geq 5$, define the invariant lattice

$$\Lambda_n = \begin{cases} 2^e\mathbb{Z}^2, & n = 2e+3, \\ 2^e\langle (1,1),(1,-1)\rangle, & n = 2e+4, \end{cases} \qquad e \geq 1,$$

and put $H_n = W/\Lambda_n$. Its order is 2^n; the full linear D_8 coordinate is retained, even if its action on the vertex quotient is not faithful. Its pair orders are $4,4,2$. It has all Boolean deletions and IP. Its third lower-central subgroup is $2\mathbb{Z}^2/\Lambda_n$, of order 2^{n-5}.

Proof. The lattice indices are respectively 2^{2e} and 2^{2e+1}, so the normal form gives the stated orders. The three linear pair products have orders four, four and two; hence the marked pair orders cannot fall in either lattice quotient.

Write $T = abcb$, $U = cbab$, translations by $(-1,0)$, $(0,1)$. The extra relators for H_n can be taken as T^{2^e}, U^{2^e} in the square case, and $T^{2^e}U^{2^e}, T^{2^e}U^{-2^e}$ in the diagonal case. After deleting any marker, both T, U have order at most two: deleting a leaves conjugates of c, deleting c leaves conjugates of a, and deleting b leaves ac, ca. Therefore every lattice relator survives every deletion. The even Coxeter relators also survive. Thus H_n has Boolean deletions and IP by UA.D.

Moreover,

$$\gamma_3(W) = 2\mathbb{Z}^2, \qquad \gamma_3(H_n) = 2\mathbb{Z}^2/\Lambda_n, \qquad |\gamma_3(H_n)| = 2^{n-5}. \tag{S2}$$

Indeed reduction modulo $2\mathbb{Z}^2$ makes the adjacent squares central; conversely $[[a,b],c]$ and $[[b,c],a]$ give the two doubled unit translations, up to signs. Thus H_5 is class two of order 32. The squared

translations are not always independent cyclic subgroups: in H_6 they coincide and have order two. Formula (S2) uses a lattice index and does not require independence of those cyclic subgroups. □

Lemma UA.SquareRecognition (all finite two-group square markings). Every finite two-group string C-group of type $\{4,4\}$ and order 2^n is marked-isomorphic to H_n; necessarily $n \geq 5$.

Proof. Its two adjacent dihedral parabolics have orders eight and intersect in order two, so their product already has 32 elements. Thus $n \geq 5$. The complete affine presentation of UA.Affine maps onto the given group G; write N for its kernel. IP gives irredundancy, and UA.Phi identifies G/G' with the elementary abelian group on its three markers. Summing those three binary coordinates gives a character sending each reflection to the nonidentity element of C_2. In the affine group this character is orientation parity. Hence every element of N preserves orientation.

The kernel contains no nonidentity torsion. A finite-order plane isometry has a fixed point, obtained for example by averaging the orbit of any point. The square reflection group moves that point into a closed fundamental right triangle. The stabilizer of a point in that triangle is trivial in the interior, generated by its wall reflection on an open edge, or the finite dihedral group generated by the two walls meeting at a vertex. These stabilizers are conjugates of the proper marked parabolics. Their maps into G are injective by its exact pair orders and IP, so none contains a nonidentity element of N.

An orientation-preserving plane isometry is a translation or a rotation. The linear part here has finite order in D_8, so a nontranslation rotation has finite order. Thus all elements of N are translations. Since N is normal of finite index, it is a D_8-invariant sublattice of $\mathbb{Z}^2$, of index 2^{n-3}. Under $\mathbb{Z}^2 = \mathbb{Z}[i]$, invariance under a quarter-turn makes it an ideal. The Euclidean algorithm in $\mathbb{Z}[i]$ makes it principal; its generator has norm a power of two. As 2 is a unit times $(1+i)^2$, unique factorization shows that this generator is a unit times a power of $1+i$. Even powers give the square lattice and odd powers the diagonal lattice in UA.Square. There is one such ideal at each two-power index. Therefore $N = \Lambda_n$ and the marked quotient is H_n. □

Proposition UA.SquareDecomposition (the two square decompositions). For every integer $q \geq 1$, let $M_2(q)$ and $M_1(q)$ be the groups obtained by adjoining respectively $(bcba)^q = 1$ and $(cba)^{2q} = 1$ to the complete presentation (S1). They are the square and diagonal lattice quotients, and have the decompositions (S3) and (S4) below. Here D_{2q} is the dihedral group of order $2q$, including $D_2 = C_2$ and $D_4 = C_2^2$.

Proof. Here bcba is translation by $(1,0)$, and $(cba)^2$ is translation by $(1,1)$. Their signed-permutation orbits therefore generate exactly the square and diagonal lattices in (S3)–(S4). This identifies those complete presentations with the indicated lattice quotients.

In the square case, $\langle a, c^b \rangle$ acts on the first coordinate and $\langle a^b, c \rangle$ on the second. Their translations have order q; each is D_{2q}, since its reflection has nonidentity linear part, also when q=1. They commute and intersect trivially. The reflection b interchanges the two factors and lies outside their product. They and b generate all three original generators. Hence

$$W/q\mathbb{Z}^2 \cong (D_{2q} \times D_{2q}) \rtimes C_2. \tag{S3}$$

In the diagonal case, put $A_0 = \langle b^a, b^c \rangle$ and $B_0 = \langle b, b^{ac} \rangle$. Their reflecting axes are respectively $x+y = 0, 1$ and $x-y = 0, -1$. The first factor has translations along $(1,1)$, the second along $(1,-1)$, of order q in each case. Each reflection retains its nonidentity linear part, so the two factors have order 2q also at q=1. All cross-generators commute. Their linear parts show that an element of their intersection must be a translation; the independence of the two diagonal translations modulo the defining lattice then makes that intersection trivial. Thus $A_0B_0 \cong D_{2q} \times D_{2q}$.

Both a and c exchange the two factors by conjugation, so normalize their product. The two individual reflections a,c do not belong to the product by their linear parts. The only remaining possible nontrivial intersection with $\langle a, c \rangle$ is ac. Every translation part of an element of A_0B_0 with linear part -1 has coordinates of the same parity, whereas ac has translation part $(0,1)$. The diagonal lattice preserves this parity condition, so ac is also outside the product. Finally b already lies in B_0, so the two

factors and $\langle a, c\rangle$ generate the full group. Consequently

$$W/q\langle(1,1),(1,-1)\rangle \cong (D_{2q} \times D_{2q}) \rtimes (C_2 \times C_2). \tag{S4}$$

These arguments specify the actions by the original affine generators and prove the decompositions, rather than only equality of orders. □

10 Changing the orders on selected pairs

Theorem UA.Edge (dihedral attachments). Let $B = \langle b_i : i \in I\rangle$ be marked by nonidentity involutions and have every Boolean deletion. Select a family $\mathcal{E}$ of distinct pairs $S = \{i, j\}$ for which the product order $q_S = |b_i b_j|$ is finite and even. For each selected pair choose a positive multiple m_S of q_S, and attach the marked dihedral quotient

$$D_{2m_S} \longrightarrow B_S = D_{2q_S}.$$

The full compatible product of UA.SK is generated by its prescribed marker tuples. It has the intersection property and all Boolean deletions. Every selected pair has product order m_S, and every unselected pair retains its original product order. If B is finite, then for every $J \subseteq I$,

$$|H_J| = |B_J| \prod_{\substack{S \in \mathcal{E} \\ S \subseteq J}} \frac{m_S}{q_S}. \tag{E1}$$

If the base marking gives a regular hypertope, so does the constructed marking, and its full type-preserving automorphism group is H.

Proof. A group generated by two nonidentity involutions whose product has finite order q_S is the dihedral group D_{2q_S}; the normal form alternates between powers of the product and a single reflection. The indicated quotient exists because $q_S \mid m_S$. It is injective on each proper local parabolic, which is either trivial or generated by one reflection. UA.D supplies IP in the base, and all input deletions are part of the hypothesis. Thus UA.SK applies. The pair supports form an antichain, so UA.SK.Local identifies every selected pair subgroup with its full local dihedral group. UA.SK.Delete gives the asserted deletions. The kernel on a selected pair has order m_S/q_S, and (E1) is UA.SK's order formula.

For an unselected pair, each extra coordinate of its product is a reflection or the identity, because that coordinate's support cannot contain both markers. If the base product has finite order q, its image under a deletion keeping just one of the two markers has order two, so q is even. The extra coordinates therefore do not enlarge its order. If the base product has infinite order, its lift has infinite order by projection. This proves the assertion for every unselected pair.

All local markers are dihedral reflections, hence involutions. Under the final hypothesis, UA.SK.Geometry gives the geometric conclusion and identifies the full type-preserving automorphism group. □

11 Amalgams and primary torsion

Fix injective homomorphisms $i_A : C \to A$ and $i_B : C \to B$. Let $A * B$ be the free product and let

$$P = (A * B)/\langle\langle i_A(c) i_B(c)^{-1} : c \in C\rangle\rangle. \tag{A1}$$

We use the amalgam normal-form theorem in the following precise form: the canonical maps of A and B into P are injective; an alternating reduced word with factors outside the common copy of C has nonzero reduced length and is nonidentity; every element is conjugate either into a factor or to a cyclically reduced alternating word of length at least two, whose positive powers remain reduced with proportionally increasing length. This is a retained classical theorem.

Theorem UA.Amalgam (universal property and torsion). The group P in (A1), with its canonical maps j_A, j_B, is a pushout of $A \leftarrow C \rightarrow B$ in the category of groups: for every group D and homomorphisms $f_A : A \to D$, $f_B : B \to D$ satisfying $f_A i_A = f_B i_B$, there is a unique homomorphism $f : P \to D$ with $f j_A = f_A$ and $f j_B = f_B$.

Let p be a prime. Say that a group has p-primary finite torsion when every finite-order element has p-power order. This property passes to subgroups, to extensions with finite p-group quotient, and to amalgamated free products of groups having the property. If $i_A(C) \neq A$ and $i_B(C) \neq B$, then P is infinite. If A and B are finitely generated, then so is P.

Proof. The free-product universal property gives a unique homomorphism $A * B \to D$ extending f_A, f_B. Agreement on C puts every defining relator in its kernel, so it factors through P. The images of A and B generate P; hence this factorization is unique. The equality $j_A i_A = j_B i_B$ follows from the defining relations, completing the pushout property.

The torsion assertion for subgroups is immediate. For an extension $1 \to K \to E \to R \to 1$, with R a finite p-group and K having p-primary finite torsion, take any finite cyclic subgroup $\langle x \rangle$ of E. Its intersection with K is a cyclic p-group, and its image in R is a p-group. The order formula for this restricted exact sequence makes $|x|$ a power of p.

For an amalgam, the normal-form theorem places a torsion element, up to conjugacy, in a factor: the alternative cyclically reduced word has powers of unbounded reduced length and cannot have finite order. The factor hypothesis therefore proves the torsion assertion. If the common subgroup is proper in both factors, choose $a \in A \setminus i_A(C)$ and $b \in B \setminus i_B(C)$. The words $(ab)^n$ are reduced of length $2n$, so are all distinct; this proves infinitude. Finally, the images of finite generating sets of A and B generate their free product and its quotient P, proving finite generation. □

The simultaneous fiber product of UA.SK has its displayed compatibility equations and its proved generation theorem. It is not the pushout in UA.Amalgam. A graph of proof dependencies is neither of these group constructions.

12 Retained inputs, applications and attribution

The proofs use the elementary subgroup, quotient and isomorphism theorems. Square recognition also uses the Euclidean algorithm and unique factorization in $\mathbb{Z}[i]$, and the elementary point stabilizers in the square reflection tiling described in its proof. The further classical inputs are explicitly limited: Burnside's basis theorem for finite p-groups; their nilpotence; generation of the lower-central graded Lie ring by degree one and its multilinear commutator rules; and the precise amalgam normal-form theorem stated in the amalgam section. The Frattini, forest, finite-quotient and torsion deductions are proved in full above. No finite group classification, atlas census or numerical search result supplies a premise to these theorems.

For applications to abstract regular polytopes we retain the following standard correspondence [MS02, Section 2E]. A group generated by an ordered list of nonidentity involutions, with the string commutation rules and IP, is the full automorphism group of a regular abstract polytope with that distinguished marking. Its rank is the length of the list, its type consists of the adjacent product orders, and its number of flags is the group order. A generator-preserving epimorphism between such groups induces a rank- and adjacency-preserving covering of their associated polytopes. These are geometric input theorems, not new conclusions of the present paper. By contrast, the coset-geometric recognition and coordinate-kernel Helly statements needed for the hypertope branch are proved in the incidence section with their full hypotheses.

The supported-extension construction supplies a common proof interface for the two-power constructions studied in [HFL20, HFL19a, HFL19b, CFHS25]. The presentation-recognition theorem supplies the additional normality and exact-base conditions needed to identify a compatible product with a complete universal presentation; local construction alone does not do this. The forest quotient gives the canonical group underlying the main quotient and cover conclusions of [HS26], and the

finite-kernel parameter space gives the exhaustive part of the order-1024 reconstruction associated with [GLP18]. Their exact source statements, literal corrections, numerical certificates and named-object identifications belong to the separate applications.

The supported-kernel induction, forest quotient and affine models are reused complete proofs. The normal-kernel recognition theorem replaces repeated finite-order comparisons, and the mix criterion separates joint coordinate detection from generation of a fiber product. Every application still proves its local defining relations, normality and base identifications. Original source results and borrowed geometric theorems keep their attribution.

Existing Lean certificates cover declared components of the kernel, supported-extension and Helly arguments. They do not presently formalize this whole paper: in particular, the full forest quotient proof, the affine model and the geometric recognition/application interfaces are not all connected in Lean. Their written proofs above remain part of the mathematical obligation. The accompanying certificate scope distinguishes inherited checked declarations from any freshly checked application components.

References

[MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. The regular-polytope and covering correspondences are retained at the scope stated above.

[HFL20] D.-D. Hou, Y.-Q. Feng and D. Leemans, *On regular polytopes of 2-power order*, revised author manuscript dated 8 November 2019; published in *Discrete & Computational Geometry* 64 (2020), 339–346. DOI: 10.1007/s00454-019-00119-5.

[HFL19a] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-polytopes of order* 2^n, arXiv:1802.08947v2, 22 January 2019; published in *Journal of Group Theory* 22 (2019), 579–616.

[HFL19b] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019; published in *Discrete Mathematics* 342 (2019), 1857–1863.

[HS26] D.-D. Hou and E. Schulte, *String C-groups of 2-power order project onto a common string C-group*, arXiv:2605.20691v1, PDF dated 21 May 2026. The specified source edition is used without asserting that every auxiliary claim is correct.

[CFHS25] G. Cunningham, Y.-Q. Feng, D.-D. Hou and E. Schulte, *Symmetric polytopes whose automorphism groups are 2-groups*, arXiv:2512.15511v1, PDF dated 18 December 2025.

[GLP18] Y. Gomi, M. L. Loyola and M. L. A. N. De Las Peñas, *String C-groups of order 1024*, *Contributions to Discrete Mathematics* 13(1) (2018), 1–22. DOI: 10.55016/ojs/cdm.v13i1.62475.